

Smart Campus Energy Intelligence System

An ML-Powered Framework for Energy Monitoring, Anomaly Detection, Demand Forecasting, and Sustainability Optimization in University Campuses

Raju Khade¹, Atharva Joshi², Nikhil Patil³, Ganesh Raghtate⁴

1,2,3,4 Department of Electronics and Computer Science Engineering, Vidyalkar Institute of Technology, Mumbai, India

ABSTRACT

One of the biggest consumers of electricity in the urban environment is large educational campuses—but even in today's day and age, most of them still rely on traditional methods of manually recorded electricity consumption statistics, without any form of actual visibility or predictive analysis being provided. In response to this problem, we created the Smart Campus Energy Intelligence System (SCEIS)—an end-to-end machine learning solution for turning time series data from smart meters into energy intelligence in a multi-building campus environment.

The system employs time series analysis in its six-step pipeline: time series processing and feature engineering; exploratory data analysis (EDA); unsupervised anomaly detection with Isolation Forest algorithm; supervised learning with electricity consumption predictions using Random Forest and Linear Regression; 24-hour ahead energy demand prediction; and sustainable analysis of electricity costs, carbon footprint, and building efficiencies.

Tests carried out on a dataset containing 33,271 hourly data points with 15 attributes demonstrate that Random Forest achieves an R^2 value close to 0.89, significantly better than Linear Regression ($R^2 \approx 0.63$). The anomaly detector correctly identifies 3% of data points as abnormal consumption events, mainly during nighttime hours. According to SCEIS, energy savings could reach 30% annually by leveraging automation during off-peak hours, resulting in a yearly saving of around ₹17.85 Lakh. This research demonstrates that responsible AI usage can help achieve UN Sustainable Development Goals SDG 7 and SDG 13.

Keywords: *Smart Campus, Energy Consumption, Artificial Intelligence, Isolation Forest, Random Forest, Anomaly Detection, Demand Forecasting, Sustainability, IoT Data Analytics, Carbon Dioxide Emission Reduction, Gradio Dashboard*

I. INTRODUCTION

The rising cost of electricity and the growing climate crisis mean that educational institutions globally have had to rethink their energy usage patterns. Engineering schools in India confront a double threat, as they need to economize on their expenditures and cut down their carbon footprints due to the reporting of emissions and the government's pledge towards achieving net zero by 2070. The cost of electricity in commercial establishments in Maharashtra state now stands at around ₹6–9 per kWh, making energy an indispensable factor in campus accounting.

An Indian engineering campus consists of a wide variety of buildings, including lecture halls, computer laboratories, administration blocks, server rooms, dormitories, libraries, and recreational centers, each having a unique footprint regarding energy use, depending on factors such as occupancy rates, equipment usage, and operating hours. Controlling energy usage in such a varied environment without intelligent data management becomes exceedingly difficult.

Standard approaches to energy management on campuses depend entirely on monthly manual metering and reactive problem resolution. If there is a problem with the air conditioning system, or if equipment was accidentally left running throughout the night, this will not be detected until the end of the billing period. The lack of a predictive element means that it will not be possible to forecast energy use or arrange for Time-of-Day (ToD) rate structures.

The three major technological developments that create an interesting possibility include: the increasing use of IoT-based smart meters; the availability of open source machine learning software including scikit-learn, pandas, and Gradio; and the ability to access cloud computing technology including Google Colab. SCEIS integrates all the enabling technologies into one deployable command addressing five primary purposes: anomaly detection, prediction of future demand, measuring waste, carbon footprint calculation, and providing a user-friendly interface.

A. Scope and Key Contributions

- Complete ML pipeline for campus energy intelligence, executable on any CSV file with five default columns.
- Comparison between Random Forest and Linear Regression for predicting multi-building energy consumption, resulting in a 41% reduction in Mean Absolute Error.
- Concurrent implementation of Isolation Forest algorithm with temporal and occupancy knowledge to detect multivariate anomalies.
- Recursive 24-hour forecasting mechanism using lagged and mean features, with accuracy measured by MAE of ~ 7.3 kWh/hour at campus level.
- Reporting module for sustainability analysis calculating cost and CO₂ emissions based on BEE annual auditing guidelines.
- Interactive Gradio web dashboard with no-code capability for non-technical users.

II. LITERATURE REVIEW

A. Data-Driven Building Energy Prediction

According to Amasyali and El-Gohary (2018), among 96 data-driven building energy prediction papers published between 2010 and 2017, ensemble trees can achieve comparable or even better prediction performance than ANNs and SVMs if data is sparse and the number of dimensions is high. Occupancy, temperature, and time-of-day were found to be the top-three important predictors of building energy consumption.

It is a known fact from the smart grid area that the importance of lags and rolling window statistics cannot be understated when predicting energy consumption. The feature Lag1 emerges as one of the dominating factors in predictive modeling. Our choice of features (Lag1, Lag2, Lag6, RollMean6) follows this known pattern with the addition of building encoding.

B. Anomaly Detection in Energy Systems

Time-series analysis for energy consumption poses peculiar problems for outlier detection. Energy consumption data are characterized by non-stationarity, seasonal components, right-skewed distributions, and multivariate outliers. Isolation Forest was proposed by Liu, Ting, and Zhou (2008) as a more computationally efficient way of detecting outliers, with linear scalability $O(n)$ and no assumptions on data distribution.

C. Deep Reinforcement Learning and Advanced Forecasting

Mocanu et al. (2018) confirm that DRL can provide significant benefits for HVAC scheduling via model-free policy learning. Nonetheless, the high cost of data, training complexity, and lack of interpretability make DRL unsuitable for academic research with limited resources. This explains why SCEIS utilizes interpretable ensemble techniques, prioritizing reproducibility and alignment with ethical AI principles.

D. Smart Campus Energy Management Platforms

Unlike commercial smart campus solutions that incorporate real-time streaming from IoT sensors and SCADA systems but cost considerable sums of money, the SCEIS system offers equivalent analytics for free. The open-source and no-cost nature of the system proves that intelligent energy management does not require expensive investments, making it viable for budget-constrained institutions in developing countries.

E. Ethical Dimensions of AI in Energy

Deploying AI in institutional resource management raises legitimate ethical concerns. Transparency is paramount: energy managers making capital investment decisions on the basis of model outputs need comprehensible justifications. SCEIS addresses this through Random Forest feature-importance visualization. An additional concern is occupant welfare: aggressive automated reductions in HVAC operation risk discomfort or harm to building users. The off-peak waste identification logic incorporates occupancy-aware decision boundaries to guard against this.

III. PROBLEM STATEMENT AND RESEARCH OBJECTIVES

University campuses in India and comparable emerging-economy environments face five critical gaps in energy management:

1. Lack of Visibility: Monthly meter readings provide no per-hour or per-building breakdown, making investigation of high bills difficult.
2. Reactive Governance: Equipment faults or unauthorized overnight usage are discovered only after billing, rendering timely remedy impossible.
3. Lack of Forecasting: No mechanisms exist to forecast demand peaks or support Time-of-Day pricing negotiations.

4. Hidden Carbon Costs: No calculation and reporting mechanisms are available for institutional carbon cost of energy decisions.
5. Unapproachable Infrastructure: Modern energy control solutions require specialized software and hardware most universities cannot afford.

In response, this study seeks to achieve five goals:

- Design and implement an automated anomaly detection system that identifies abnormal events without manual data analysis.
- Create a supervised regression algorithm predicting per-building energy consumption, benchmarked against a linear regression baseline.
- Create a recursive energy demand forecast algorithm operating on a 24-hour rolling basis for campus-wide load management.
- Develop a sustainability insights system measuring costs, CO₂ emissions, off-peak wastage, and automation savings potential.
- Make all analytics available through a user-friendly web dashboard requiring no programming skills.

IV. DATASET DESCRIPTION AND CHARACTERISTICS

A. Dataset Overview

The dataset, named 'Smart Campus Energy Enhanced,' is an artificially generated time series dataset simulating real-life energy consumption on a multi-building university campus. It consists of 33,271 hourly observations and 15 features, covering six months of data across multiple building types including academic, residential, and administrative blocks.

B. Feature Schema

#	Feature Name	Type	Description
1	Date	Datetime	Observation timestamp (hourly)
2	Building_Name	Categorical	Unique building identifier
3	Building_Type	Categorical	Academic / Residential / Admin
4	Energy_Consumption_kWh	Float	Total electricity consumed (target)
5	Temperature_C	Float	Ambient temperature

#	Feature Name	Type	Description
			(°C)
6	Humidity_Pct	Float	Relative humidity (%)
7	Occupancy_Count	Float	Number of occupants
8	Device_Count	Integer	Number of active appliances
9	Holiday	Integer	Holiday flag (0=No, 1=Yes)
10	Power_Factor	Float	Power usage efficiency ratio
11	CO ₂ _kg	Float	CO ₂ emissions (derived)
12	Renewable_Energy_kWh	Float	Solar energy offset

TABLE I: Dataset Feature Schema

C. Key Dataset Characteristics

The hourly granularity of the dataset enables inclusion of lag and rolling window features that capture time dependency. The dataset contains both numerical and categorical features. Linear interpolation is used for Temperature_C since it varies continuously; median imputation is used for Occupancy_Count due to its right-skewed distribution. Energy_Consumption_kWh is right-skewed and occasionally exhibits spikes due to equipment malfunction and simultaneous operation of multiple systems.

V. METHODOLOGY

A. System Architecture Overview

The SCEIS pipeline consists of six modules, each assigned to a corresponding tab in the Gradio interface. Pre-processing occurs only once during ingestion, after which the DataFrame is passed through to all subsequent modules, ensuring consistent analysis.

Module	Algorithm	Primary Output
1. Data Preview	Statistical + pandas	Row preview + metadata
2. EDA	Aggregation + Matplotlib	6 analytical charts
3. Anomaly Detection	Isolation Forest (3%)	Anomaly flags + viz
4. ML	Random Forest +	R ² , MAE, RMSE

Module	Algorithm	Primary Output
Prediction	LR	
5. Forecasting	RF Rolling Recursive	24-hour forecast
6. Sustainability	Rule-based statistical	8 recommendations

TABLE II: SCEIS Pipeline Module Summary

B. Data Pre-processing and Feature Engineering

Sensor data undergoes seven stages of pre-processing before being fed into any analysis module:

Step 1 – Data Validation and Datetime Conversion: Five columns are verified for existence, dates are parsed, and the DataFrame is sorted to construct lags correctly.

Step 2 – Temperature Interpolation: Missing Temperature_C values are imputed using linear interpolation as it is a physically continuous time series.

Step 3 – Occupancy Imputation: Missing Occupancy_Count values are imputed using the column median to handle right-skewed distribution.

Step 4 – Temporal Feature Creation: Hour (0–23), DayOfWeek (0 = Monday), and IsWeekend (Boolean) are derived from the DateTime index.

Step 5 – Lag Features: Lag1 is calculated per building group by offsetting Energy_Consumption_kWh by one timestep — the single best predictive feature (~38% importance).

Step 6 – Rolling Mean Feature: RollMean6, a 6-hour rolling mean per building group, is created to capture short-term trends.

Step 7 – Derived Economic Metrics: Cost_INR = Energy $\times$ ₹7/kWh (Maharashtra commercial tariff) and CO₂_kg = Energy $\times$ 0.82 kg/kWh (CEA 2023 emission factor).

C. Anomaly Detection: Isolation Forest

An Isolation Forest is implemented to detect multivariate anomalies without assuming data distributions. The model operates using ensembles of isolation trees (iTrees), constructed by recursive partitioning of the feature space with random split points. The contamination level is set to 0.03 (3% anomaly budget). Input features: Energy_Consumption_kWh, Occupancy_Count, Temperature_C, and Hour. Labels +1 (normal) and -1 (anomaly) are stored in a new Anomaly column.

D. Predictive Modelling: Random Forest vs. Linear Regression

Two regression models are deployed: Random Forest Regression and Ordinary Least Squares Linear Regression. Both are tested using an 80/20 random train-test split (random_state=42). Random Forest parameters: n_estimators=80, max_depth=12, n_jobs=-1 for parallel learning. Eight predictor variables are used: Hour, DayOfWeek, IsWeekend, Temperature_C, Occupancy_Count, Lag1, RollMean6, and Building_Code. Linear regression serves as an interpretable baseline. Model accuracy is measured using R², MAE, and RMSE.

E. Demand Forecasting: Rolling Recursive Architecture

The forecasting model predicts aggregated campus energy consumption 24 hours ahead. Campus aggregates are created by summing all building outputs at each timestamp. A custom Random Forest Regressor (n_estimators=60, max_depth=10) is fitted using an 85%–15% time-series split without data leakage. At each of 24 subsequent timesteps, the model inputs the most recent lagged values (Lag1, Lag2, Lag6), rolling mean (RollMean6), and adds the forecast to the history buffer for the next round.

F. Sustainability Insights Module

Off-peak Waste Detection: Records meeting all three conditions are flagged: (a) not weekend; (b) outside standard business hours (08:00–18:00); and (c) consumption above the 70th percentile. A 30% reduction factor is applied to estimate automation savings.

Facility Efficiency Scoring: Buildings are scored 0–100 based on energy intensity vs. occupancy, off-peak ratio, and renewable offset. Scores ≥ 70 = green, 50–70 = amber, < 50 = red, enabling facility managers to prioritize attention.

Carbon Footprint Calculation: CO₂ emissions are calculated using India's 2023 grid emission factor (0.82 kg/kWh, CEA). Equivalent trees required annually to offset emissions are also computed (21 kg CO₂ per tree).

VI. IMPLEMENTATION DETAILS

A. Technology Stack

Component	Technology	Purpose
Language	Python 3.10+	Core implementation
Data Processing	pandas + NumPy	Feature engineering
Machine Learning	scikit-learn 1.4+	RF, LR, Isolation Forest
Visualisation	Matplotlib 3.8+	Charts and plots
UI Framework	Gradio 4.x	Web dashboard
Deployment	Google Colab	Zero-cost cloud execution

TABLE III: Technology Stack

B. Key Code Segments

Pre-processing Function:

```
def preprocess(df):
    df['Date'] = pd.to_datetime(df['Date'])
    df['Lag1'] =
df.groupby('Building_Name')
    ['Energy_Consumption_kWh'].shift(1)
    df['Cost_INR'] = df['Energy_...'] * 7
    df['CO2_kg'] = df['Energy_...'] * 0.82
```

Isolation Forest Implementation:

```
iso = IsolationForest(contamination=0.03,
                      random_state=42)
feats = df[['Energy_Consumption_kWh',
            'Occupancy_Count', 'Hour']]
df['Anomaly'] = iso.fit_predict(feats)==-1
```

VII. EXPERIMENTAL RESULTS AND ANALYSIS

A. ML Model Performance Comparison

Model	R ²	MAE (kWh)	RMSE (kWh)
Linear Regression	~0.63	~12.4	~18.7
Random Forest	~0.89	~5.8	~9.2
Improvement (Δ)	+41.3%	-53.2%	-50.8%

TABLE IV: Model Performance Comparison

Random Forest outperforms Linear Regression on all three criteria. The 26-point increase in R² represents the additional proportion of consumption variation explained by the non-linear model. The 53.2% decrease in MAE from 12.4 kWh to 5.8 kWh per estimate is operationally significant: facility managers allocating HVAC loads can tolerate 5.8 kWh error but not 12.4 kWh. This difference arises because cooling demand is a product of temperature and occupancy — not their sum — which cannot be captured by linear regression without explicitly specified interaction terms.

B. Feature Importance Analysis

Rank	Feature	Importance	Interpretation
1	Lag1	~38%	Previous-hour energy — dominant momentum
2	RollMean6	~22%	6-hour rolling avg — trend context
3	Occupancy_Count	~14%	Direct demand driver
4	Hour	~11%	Diurnal cycle pattern
5	Temperature_C	~8%	Cooling load (non-linear)
6	Building_Code	~4%	Fixed building effects
7	DayOfWeek	~2%	Weekly pattern
8	IsWeekend	~1%	Binary weekend signal

TABLE V: Random Forest Feature Importance

The combination of Lag1 (38%) and RollMean6 (22%) accounts for 60% of prediction capability, supporting the robustness of building energy data's autoregressive nature. This finding underscores the importance of sensor reliability: without reliable lag data, forecasting accuracy degrades significantly.

C. Anomaly Detection Results

Metric	Value
Total records analysed	33,271 hourly readings
Anomalies detected	999 records (3.0%)
Average anomalous reading	891.0 kWh (vs 294.1 kWh normal)
Anomaly-to-normal ratio	3.03× normal mean consumption
Most affected building	CS Department (496 anomalies)
Peak anomaly hour	02:00–04:00 (overnight faults)

TABLE VI: Anomaly Detection Results

Anomaly clusters in the CS Department during nighttime hours can be explained by cooling systems for servers operating outside scheduled periods, or high-power computer clusters running at off-peak times. Such anomaly clusters would not be observable with monthly metering, demonstrating the clear value of SCEIS for proactive campus management.

D. Forecasting Performance

Metric	Value
Test-set MAE	~7.3 kWh/hour (campus aggregate)
Test-set RMSE	~11.8 kWh
24-hour forecasted total	~3,200 kWh (5-building campus)
Forecasted peak hour	14:00 (post-lunch AC peak)
Estimated 24-hour cost	₹22,400
Estimated 24-hour CO ₂	~2,624 kg

TABLE VII: 24-Hour Forecasting Performance

A campus-level MAE of 7.3 kWh/hr indicates a relative prediction error of only 0.23% when the campus consumes 3,200 kWh/day, falling within tolerances required by demand-side management agreements with electricity distribution utilities.

E. Sustainability Insights

Sustainability Metric	Value
Annual energy consumption	~850,000 kWh
Annual electricity cost	₹59.5 Lakh
Annual CO ₂ emissions	~697 tonnes CO ₂
Trees to offset emissions	~33,190 trees/year
Off-peak waste potential	~255,000 kWh/year
Annual savings via automation	₹17.85 Lakh
CO ₂ reduction potential	~473,000 kg/year
Estimated payback period	7–12 months

TABLE VIII: Sustainability Insights Summary

The anticipated savings of ₹17.85 Lakh per annum represent 30% of total electrical consumption costs. Taking into account IoT automation system costs of ₹5–10 Lakh for a five-building deployment, the calculated payback period is 7–12 months, representing an excellent return on investment.

VIII. ETHICAL DIMENSIONS OF THE SYSTEM

8.1 Interpretability: All predictive models in SCEIS are inherently interpretable. Random Forest importance scores are explicitly calculated and visualized. Isolation Forest anomaly scores are accompanied by underlying feature values, enabling facility managers to verify flagged records before any decision-making process.

8.2 Privacy Protection: All data in SCEIS represents building-level aggregated counts. At no point does SCEIS use personally identifiable information. Extensions utilizing individual-level movements would require specific privacy protection and data minimization policies.

8.3 Fairness and Resource Distribution: The risk of discriminatory resource allocation through automation is mitigated by the SCEIS efficiency model, which compares consumption relative to occupancy and floor space, not absolute values. This ensures a 24-hour laboratory is not unfairly penalized compared to an 8-hour administrative office.

8.4 UN Sustainable Development Goals: SCEIS operationalizes SDG 7 (Affordable and Clean Energy) by lowering energy costs and monitoring renewable energy, and SDG 13 (Climate Action) through CO₂ emission calculation and reduction planning. It demonstrates AI as a sustainability tool rather than purely a commercial one.

IX. APPLICATIONS AND FUTURE DIRECTIONS

A. Immediate Applications

- **Campus Energy Management:** Estate managers submit monthly smart meter readings for instant anomaly analysis, energy forecasting, and efficiency ranking without technical knowledge.
- **Regulatory Compliance:** CO₂ and cost figures auto-populate information needed for BEE energy audits and NAAC green campus certification.
- **Demand-Side Management:** Real-time forecasting enables peak load management contracts and Time-of-Day savings.
- **Fault Detection and Maintenance:** Off-peak anomalies help detect HVAC, UPS, and server cooling faults early.

B. Scalability Pathways

- **Cloud Hosting:** The Colab model is deployable as a 24×7 web application using Hugging Face Spaces, AWS Lambda, or Azure App Services.
- **IoT Real-time Ingestion:** CSV-based ingestion can be replaced with stream-based connections to Schneider Electric EcoStruxure or Siemens Desigo CC APIs.
- **Federated Multi-Campus Dashboard:** Adding a Campus_Name column enables federated visualization across multiple campuses.

C. Future Research Enhancements

- Deep Learning Forecasting: LSTM and Transformer models to improve long-term seasonal accuracy.
- Solar PV Integration: Net electricity calculation accounting for renewable generation and LCOE calculations.
- Conversational AI Layer: Natural language interface for estate managers via LLM API.
- Reinforcement Learning Optimization: DRL agent for real-time HVAC scheduling.
- Peer Benchmarking: Comparative energy intensity per m² per occupant across similar institutional campuses.

X. CONCLUSION

The Smart Campus Energy Intelligence System is a robust, implementable machine learning solution enabling active, intelligent governance by turning passive energy data into meaningful intelligence. Using Isolation Forest anomaly detection, Random Forest demand prediction, recursive forecasting, and rules-based sustainability assessment, SCEIS resolves all major issues plaguing campus energy management: real-time visibility problems, unaccounted anomalies, lack of forecasting ability, unknown carbon costs, and lack of accessibility for non-technical stakeholders.

Experimental findings show that Random Forest substantially outperforms the linear regression baseline ($R^2=0.89$ vs. 0.63), with MAE improved by 53.2%. The anomaly detection system tags 3% of entries as abnormal, consuming $3.03\times$ the mean electricity of normal readings. The 24-hour forecasting model produces a campus-wide MAE of 7.3 kWh, adequate for demand-side management. The sustainability module estimates annual savings of ₹17.85 Lakh using off-peak automation, with a payback period under one year.

Beyond technical merit, SCEIS demonstrates how machine learning can be used responsibly to deliver tangible social benefits. Every design decision — from using interpretable ensemble methods to building-level aggregated occupancy data, from occupancy-normalized efficiency scoring to open-source deployment — is guided by the ethical principles of transparency, privacy, fairness, and sustainability. This solution relies entirely on open-source software at zero cost, enabling immediate deployment across educational institutions throughout India and comparable developing economies.

REFERENCES

- [1] Amasyali, K. and El-Gohary, N.M. (2018). A Review of Data-Driven Building Energy Consumption Prediction Studies. *Renewable and Sustainable Energy Reviews*, 81, pp. 1192–1205.
- [2] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), pp. 5–32.
- [3] Bureau of Energy Efficiency (BEE), Government of India (2023). Energy Conservation Building Code (ECBC) for Commercial Buildings. New Delhi: Ministry of Power.
- [4] Central Electricity Authority, Government of India (2023). CO₂ Baseline Database for the Indian Power Sector, Version 18.0. New Delhi: Ministry of Power.
- [5] Gradio Team (2024). Building Machine Learning Applications with Gradio: Documentation and API Reference. <https://www.gradio.app/docs/>
- [6] Liu, F.T., Ting, K.M. and Zhou, Z.-H. (2008). Isolation Forest. In: *Proceedings of the 8th IEEE International Conference on Data Mining (ICDM)*, pp. 413–422.
- [7] McKinney, W. (2022). *Python for Data Analysis*, 3rd edn. O'Reilly Media.
- [8] Mocanu, E. et al. (2018). On-line Building Energy Optimization Using Deep Reinforcement Learning. *IEEE Transactions on Smart Grid*, 10(4), pp. 3698–3708.
- [9] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, pp. 2825–2830.
- [10] United Nations (2015). *Transforming Our World: The 2030 Agenda for Sustainable Development*. SDG 7 & SDG 13. New York: United Nations.